moment after moment to be momentarily real—to be the momentary *now*—as the mechanism flows ever onward toward the future.

Quite the contrary. While the *perspective* of Figure 5.1 is certainly imaginary, there is convincing evidence that the spacetime loaf—the totality of spacetime, not slice by single slice—is real. A less than widely appreciated implication of Einstein's work is that special relativistic reality treats all times equally. Although the notion of *now* plays a central role in our worldview, relativity subverts our intuition once again and declares ours an egalitarian universe in which every moment is as real as any other. We brushed up against this idea in Chapter 3 while thinking about the spinning bucket in the context of special relativity. There, through indirect reasoning analogous to Newton's, we concluded that spacetime is at least enough of a something to provide the benchmark for accelerated motion. Here we take up the issue from another viewpoint and go further. We argue that every part of the spacetime loaf in Figure 5.1 exists on the same footing as every other, suggesting, as Einstein believed, that reality embraces past, present, and future *equally* and that the flow we envision bringing one section to light as another goes dark is illusory.

The Persistent Illusion of Past, Present, and Future

To understand Einstein's perspective, we need a working definition of reality, an algorithm, if you will, for determining what things exist at a given moment. Here's one common approach. When I contemplate reality—what exists at *this* moment—I picture in my mind's eye a kind of snapshot, a mental freeze-frame image of the entire universe right *now.* As I type these words, my sense of what exists right *now,* my sense of reality, amounts to a list of all those things—the tick of midnight on my kitchen clock; my cat stretched out in flight between floor and windowsill; the first ray of morning sunshine illuminating Dublin; the hubbub on the floor of the Tokyo stock exchange; the fusion of two particular hydrogen atoms in the sun; the emission of a photon from the Orion nebula; the last moment of a dying star before it collapses into a black hole—that are, at this moment, in my freeze-frame mental image. These are the things happening right *now,* so they are the things that I declare exist right *now.* Does Charlemagne exist right now? No. Does